Hochschule Fulda
University of Applied Sciences

Global Distributed Software Development - AI5015 / AI5088

Team Project Summer 2025 - Milestone 1

Fulda Einkaufszentrum

(Hochschule Fulda Internal Marketplace for Students, Staff and Faculty to Buy, Sell and Exchange Goods.)

(Dated: 08/05/2025)

Meet the team: **Team 4**

Name	**Email**	**Role**
Pavithra Purushothaman	pavithra.purushothaman@informatik.hs-fulda.de	Team Lead
Akhil Baiju	akhil.baiju@informatik.hs-fulda.de	Database Engineer
Mufidha Sulfikerali	mufidha.sulfikerali@informatik.hs-fulda.de	GitHub Master
Udit Parihar	udit.parihar@informatik.hs-fulda.de	Backend Developer
Ayush Kaushik	ayush.kaushik@informatik.hs-fulda.de	Frontend Lead
Ridhima Tigadi	ridhima.tigadi@informatik.hs-fulda.de	Backend Lead

Supervisor: Prof. Dr. Rainer Todtenhöfer

Table of Contents

1. Executive Summary

2. Personas and Main Use Cases
    2.1 Personas
    2.2 Use Cases

3. List of Main Data Items and Entities

4. Initial List of Functional Requirements

5. List of Non-Functional Requirements

6. Competitive Analysis

7. High-Level System Architecture and Technologies Used

8. Team and Roles

9. Checklist

1. Executive Summary

Motivation & Importance:

Hundreds of students arrive every semester at Hochschule Fulda in search of relatively inexpensive used furniture, books, or miscellaneous everyday items. Meanwhile, many students, staff, or faculty members are departing or decluttering and have items that are still perfectly usable. However , there are no safe, trusted, and local platforms available for the Fulda community to safely and easily exchange items. eBay, Kleinanzeigen and Facebook Marketplace are way too broad, lacking in trust, and rarely student-focused. The goal of this project was to provide a very easy online space for Fulda University members to buy, sell, or exchange goods with one another directly.

Key Functions & Services:

- **Signup/Login with Fulda Email**
  Only students, staff, and faculty with a Hochschule Fulda email can create an account.

- **Create or View Ads**
  Users can post items to sell, exchange, or request, and browse listings from others.

- **Search, Sort, and Filter**
  Easily find the right items using category filters, price range, and sorting options.

- **Admin Approval Before Posting**
  All ads are checked by an admin before going live to keep content relevant and safe.

- **My Ads Section**
  Users can view, edit, or remove their own listings from a personal dashboard.

- **Ad Translation Option**
  Listings can be translated to other languages, making the platform easier to use for international students.

- **Report a Spam Ad**
  Users can report inappropriate or misleading listings to help keep the platform clean.

- **Pickup Spot Suggestions**
  The platform recommends safe and familiar campus locations for meeting and exchanging items.

- **Seller's Known Languages**
  Listings can show which languages the seller speaks, making communication easier.

- **Ad Expiry Limit (180 Days)**
  Ads automatically expire after 180 days to keep the listings up-to-date.

- **User Expiry for Inactive Accounts**
  Accounts are cleaned up after long inactivity to keep the community active.

- **Campus-Only Community**
  Only Fulda members can use the platform, creating a more trustworthy and local environment.

- **Ads-Free Experience**
  No third-party advertisements or promoted listings, just a clean, focused marketplace.

- **Contact via Email Form**
  Interested buyers can send messages to sellers through a simple email form (no accounts shared).

Unique Value Proposition:

What really sets this platform apart is the fact that it is community centered and designed just for the Fulda community. Listings are only available to verified members of the university which keeps posts on topic and members safe. Features like suggested pickup spots, seller's known languages and automatic expiration of listings will make the experience continued and easy. This is not just an app, it's developed around the specific habits and needs of Fulda students and staff.

Team Overview:

Our team consists of passionate students from Fulda University who recognized a real need for a better solution that is campus-based. We have a variety of backgrounds with technical design and user experience skills and we've come together to solve problems and provide something simple, useful and community-oriented. This app is by Fulda, for Fulda.

2. Personas and Main Use Cases

2.1 Personas:

Persona 1: Andrew Meier – The Student Buyer (Primary Persona)

- **Characteristics:** Andrew is a second-year undergraduate at Fulda University. He's practical, budget-conscious, and seeks reliable second-hand items like bikes and other daily-use goods.
- **Goals:** Find affordable, good-quality used items from trusted campus sources; communicate quickly and safely with sellers.
- **Skills:** Comfortable using digital platforms, can navigate search filters and listing images, and communicates via email or phone.
- **Pain Points:** Limited student budget; concern about scams or poor product quality; prefers secure, campus-based exchanges.

Persona 2: Lillian Becker – The Student Seller (Primary Persona)

- **Characteristics:** A fourth-year student preparing for graduation, Lillian wants to declutter and earn extra cash by selling unneeded items like textbooks.
- **Goals:** Sell used textbooks and other student supplies quickly; ensure smooth, hassle-free transactions within campus.
- **Skills:** Familiar with uploading listings and responding to messages promptly.
- **Pain Points:** Time constraints during finals; doesn't want to deal with strangers or leave campus to complete transactions.

Persona 3: Yash Gupta – The International Student (Primary Persona)

- **Characteristics:** A new international graduate student unfamiliar with local shopping and cultural norms.
- **Goals:** Furnish his dorm room affordably and quickly; connect with sellers who understand his language or context.
- **Skills:** Tech-savvy and fluent in English; adapts easily to online buying platforms.
- **Pain Points:** Cultural and logistical unfamiliarity; limited local contacts; potential language barriers.

Persona 4: Professor Müller – The Faculty Seller (Secondary Persona)

- **Characteristics:** A senior faculty member in the Human Rights department, not particularly tech-savvy but community-oriented.
- **Goals:** Pass along gently-used office items to students or colleagues; declutter with minimal digital effort.
- **Skills:** Basic computer usage; can create listings with some assistance.

- **Pain Points:** Prefers simple, intuitive interfaces; avoids complicated steps or heavy moderation duties.

Persona 5: Mia Schuster – The Administrative Buyer (Secondary Persona)

- **Characteristics:** Works in the administrative department; seeks useful, budget-friendly items for staff spaces like break rooms.
- **Goals:** Find high-quality used appliances locally; avoid public marketplace hassles.
- **Skills:** Efficient with internal tools and online browsing.
- **Pain Points:** Avoids dealing with off-campus sellers or platforms lacking trust.

Persona 6: Lukas Brandt – The Guest User

- **Characteristics:** Lukas is an alumnus visiting the university campus. He's curious about the platform and wants to explore available listings but doesn't have a university login.
- **Goals:** View available products and services without account creation; get a general idea of item prices and availability.
- **Skills:** Comfortable browsing web platforms with basic navigation skills.
- **Pain Points:** Cannot contact sellers or save listings without a verified student/staff login; limited interaction options.

Persona 7: Admin Role – The Platform Moderator

- **Characteristics:** A designated campus administrator monitoring the platform's content and users.
- **Goals:** Keep listings appropriate and relevant; remove scams or flagged users; oversee analytics and maintenance.
- **Skills:** Strong backend and moderation experience; familiar with data dashboards and website operations.
- **Pain Points:** High volume of listings and users to review; balancing control with user autonomy.

Persona 8: Developer Role – The System Maintainer

- **Characteristics:** A technical team member responsible for implementing fixes and adding features based on admin feedback.
- **Goals:** Keep the platform bug-free and scalable; update inactive accounts; improve features as needed.
- **Skills:** Advanced in database management, code maintenance, and role assignment.

- **Pain Points:** Must rely on detailed admin reports; needs structured, updated requirements for continuous improvement.

2.2 Use Cases:

UC-1: Student Buys a Used Bike

- **Actor:** Andrew Meier (Student)
- **Goal:** Secure an affordable bike on campus.
- **Description:** Andrew browses listings, contacts the seller, arranges a meeting, and completes the transaction within the safety of campus premises.

UC-2: Student Sells Textbooks After Semester

- **Actor:** Lillian Becker (Student)
- **Goal:** Sell textbooks to other students.
- **Description:** Lillian posts listings, receives interest from a first-year student, and meets at the campus library for a smooth handover.

UC-3: International Student Furnishes Dorm Room

- **Actor:** Yash Gupta (Student)
- **Goal:** Buy essential furniture affordably.
- **Description:** Yash browses furniture listings, contacts multiple sellers, and picks up the items from different locations in the campus over a few days.

UC-4: Faculty Declutters via Listings

- **Actor:** Professor Müller (Faculty)
- **Goal:** Donate or sell unused office items.
- **Description:** Professor Müller uploads simple listings and coordinates pickups with interested students and staff.

UC-5: Staff Member Purchases an Appliance

- **Actor:** Mia Schuster (Administrative Staff)
- **Goal:** Find a cost-effective microwave.
- **Description:** Mia discovers a nearby listing from a faculty member, arranges pickup, and completes the transaction efficiently.

UC-6: Guest Views Public Listings

- **Actor:** Lukas Brandt (Guest)
- **Goal:** Explore available listings without signing up.
- **Description:** Lukas browses the platform's public listings (e.g., titles, categories, and photos). However, he cannot view full details, contact sellers, or post listings until he registers with a Fulda email ID.

UC-7: Faculty/Staff Flags a Scam User

- **Actor:** Professor Müller or Mia Schuster (Faculty/Staff)
- **Goal:** Report a suspicious user.
- **Description:** After noticing suspicious behavior, they use the "mark as scam" feature. If a user is flagged repeatedly, the system highlights the user for admin review.

UC-8: Admin Deletes Inappropriate Listings

- **Actor:** Admin Role
- **Goal:** Maintain listing quality.
- **Description:** Admin identifies inappropriate or scam content and removes it from the platform after review.

UC-9: Admin Monitors Dashboard and Reports

- **Actor:** Admin Role
- **Goal:** Track activity and maintenance needs.
- **Description:** Admin checks user and posting analytics, receives bug reports, and flags items for the developer to address.

UC-10: Developer Fixes Reported Bugs

- **Actor:** Developer Role
- **Goal:** Maintain smooth platform functionality.
- **Description:** Developer reviews the admin's report, updates backend components, and implements requested features or account changes.

3. List of Main Data Items and Entities

Below is a table defining key data items and entities used in your application. Including columns for Index, Entity/Item Name, and Description (Purpose/Usage).

Index	Entity/Item Name	Description (Purpose/Usage)
1	User	Used for authentication, authorization (Student, Staff, Faculty).
2	Item	Object listed for sale or exchange. Used in browsing and searching.
3	Message	In-platform communication between Buyers and Sellers.
4	Category	Helps users to browse and filter items based on their needs.
5	Report	Allows Admins to check and review listings and user behavior.
6	Approval Queue	Ensures Admin approves before listings become public.
7	Media Asset	Enriches item visibility with visual content.
8	Login Session	Tracks user activity and helps enforce usage limits.

4. Initial List of Functional Requirements

1. **User Registration and Login**
   The system shall allow users to register and log in using university email addresses to ensure campus-only access.

2. **Campus Verification**
   Verify users' university affiliation through email domain or student ID to ensure campus-only access.

3. **Product Browsing and Search**
   Users shall be able to browse and search for available products using filters like category, price, location.

4. **Create New Product Listings**
   Users shall be able to create new listings by adding images, descriptions, price, location, and whether the item is for sale or swap.

5. **Time-Limited Listings and User Access**
   Users can set product listing expiry up to 180 days; expired listings are auto-archived unless renewed. User accounts (e.g., guests) expire after 180 days unless extended by admin.

6. **Edit or Delete Listings**
   Enable sellers to update or remove their listings at any time before the item is sold or expired.

7. **View Product Details**
   Users shall be able to view detailed information about a product, including description, price, images, seller info, and location.

8. **Email-based Buyer-Seller communication**
   If users have queries related to a purchase or sale, the system shall trigger an email to the respective buyer or seller to initiate communication.

9. **Swap Offers**
   Users shall be able to propose swaps (item-for-item deals) instead of paying with money.

10. **Buy Products Directly**
    Enable students to purchase items via a simple checkout process within the platform.

11. **Role-Based Access Control**
    System distinguishes between students, faculty, guests, and admins with appropriate feature access.

12. **Trusted Seller Verification System**
    System issues badges to verified, high-rated sellers to help buyers find reliable sources.

13. **Interactive Campus Map for Item Exchanges**
    Users can select campus pickup points visually when setting up or accepting a trade.

14. **Pickup or Meetup Coordination**
    The system shall suggest nearby pickup locations on campus and allow users to select or propose meeting points.

15. **Admin Dashboard for Moderation**
    Admins shall be able to monitor listings, user activity, and flag or remove inappropriate content.

16. **User Profile and Transaction History**
    Each user shall have a profile page displaying their listings, purchases, swaps, group carts, and ratings.

17. **Bilingual Interface Support**
    The system shall support both German and English languages, allowing users to switch between them and view all interface elements and product listings accordingly.

18. **Display Seller's Known Languages**
    The system shall allow sellers to indicate the language they prefer, i.e. German or English, which will be displayed on their profile or listing to help buyers choose an appropriate communication language.

19. **Make an Offer Functionality**
    Users shall be able to send a custom price offer to the seller (or vice versa) for listed products, which the recipient can accept, reject, or counter.

20. **Report Spam or Inappropriate Content**
    Users shall be able to report listings or profiles they suspect to be spam, or inappropriate; reported content will be flagged for review.

5. List of Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions must render well on mobile devices
4. Data shall be stored in the database on the team's deployment cloud server.
5. No more than 50 concurrent users shall be accessing the application at any time
6. Privacy of users shall be protected and all privacy policies will be appropriately communicated to the users.

7. The language used shall be English (no localization needed)
8. Application shall be very easy to use and intuitive
9. Application should follow established architecture patterns
10. Application code and its repository shall be easy to inspect and maintain
11. Google analytics shall be used (optional for Fulda teams)
12. No e-mail clients shall be allowed.
13. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
14. Site security: basic best practices shall be applied (as covered in the class) for main data items
15. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today
16. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development
17. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Summer 2025 For Demonstration Only" at the top of the WWW page. (Important so as to not confuse this with a real application).

6. Competitive Analysis

6.1 Competitive Analysis Table:

Below is a table comparing key features of competitor products versus your planned product.

Feature	**eBay**	**Kleinanzeigen**	**Facebook Marketplace**	**Fulda Einkaufszentrum**
Signup/Login (Fulda email only)	✅	✅	✅	✅
Search, Sort, Filter	✅	✅	✅	✅
Create/View Ad	✅	✅	✅	✅
Approval by Admin (Manual moderation)	❌	❌	❌	✅
Ad Listing Screen	✅	✅	✅	✅

My Ads (User profile)	✅	✅	✅	✅
Admin Dashboard / Moderation Tools	❌	❌	❌	✅
Translate Ad Content	❌	❌	❌	✅
Chat support	✅	✅	✅	❌(Mail form)
Pickup Spot Recommendations	❌	❌	❌	✅
Report Spam	✅	✅	✅	✅
Show Seller's Known Languages	❌	❌	❌	✅
Ad Time Limit (Max 180 days)	❌	❌	❌	✅
User Expiry (Auto-remove after 180 days)	❌	❌	❌	✅
Campus-Only Community (Secure, local access)	❌	❌	❌	✅
Ads Free (No third-party ads/boosts)	❌	❌	❌	✅

6.2 Analysis Summary:

In contrast to widely used platforms such as eBay, Kleinanzeigen and Facebook Marketplace, our application is designed from the ground up for the Fulda University community. All of these platforms enable users to post and browse items, but our product was designed to offer a safer and more focused platform. Features like admin approval for posts, campus-only access, pickup location suggestions and ad translations, have all been built into our application 's features to ensure Fulda students and staff use and trust the application. Also, the fact we have a button to indicate which languages a seller can speak, will hopefully allow users once again break the communication barrier. The design of our application ensures every post as well as a user interface is free of ads, so users can focus on what matters to them. While we don't have real time chat at the moment, reaching out to a seller through email keeps things simple while protecting a user's privacy. Overall this is not just another marketplace , it is a product intended and deeply thought out to help build a platform for the campus community by using features designed from real campus needs.

7. High-Level System Architecture and Technologies Used

Technologies

• Server Host: AWS EC2 t2.micro (1 vCPU, 1 GB RAM)

• Operating System: Ubuntu 22.04 LTS

• Database: MySQL 8.0

• Web Server: NGINX 1.22

• Server-Side Language: Python 3.10

• Web Framework: Flask 2.3

• Frontend Library: React 18

• Cloud Storage: AWS S3

• IDE: VSCode

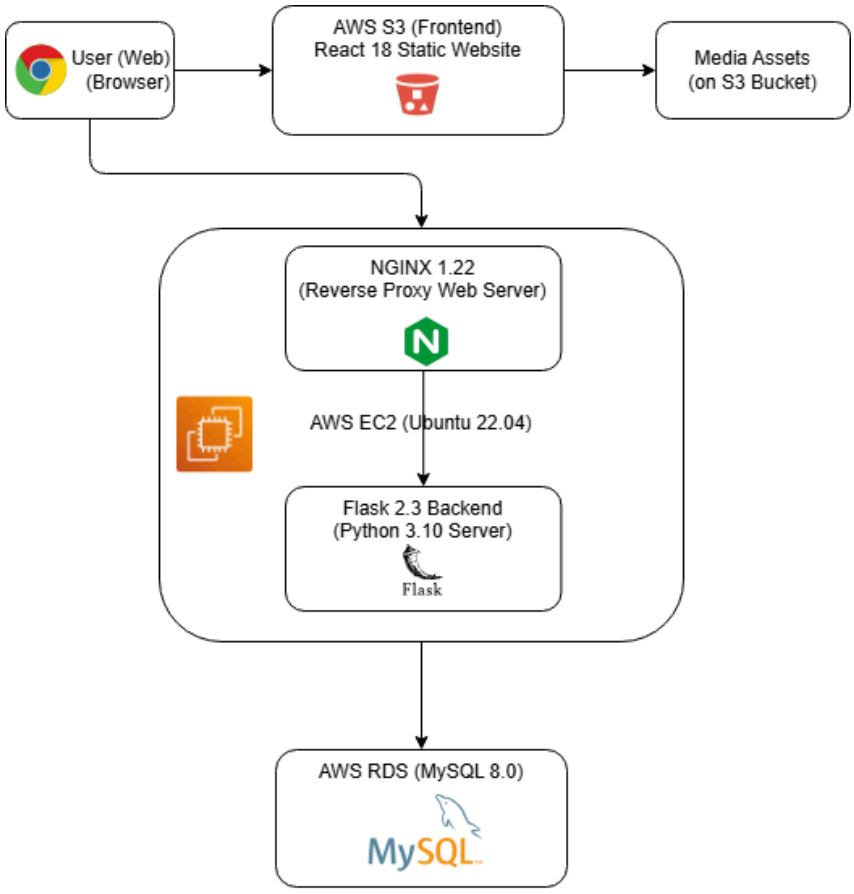


Fig: Web App Deployment Architecture

Deployment Information

- Supported Browsers: Chrome 130+, Firefox 132+, Edge, Safari, Brave (not IE)
- Deployment Method: Automated via GitHub Actions.
- How It Works:
    - The workflow is triggered on every push to the main branch.
    - It checks out the source code from the repository.
    - AWS credentials are securely configured using GitHub Secrets.
    - The files are synchronized to an AWS S3 bucket using the AWS CLI.
    - Any files removed from the repository are also deleted from the S3 bucket to keep it in sync.
- Hosting: The static website is served directly from the configured S3 bucket endpoint.

8. Team and Roles

Team's roles and responsibilities of this project.

Name	Role	Responsibilities
Pavithra Purushothaman	Team Lead	Keep the team focussed on the project goals and to communicate with CTO/CEO
Akhil Baiju	Database Engineer	Maintains and organises project database
Mufidha Sulfikerali	GitHub Master	Organizes Git and makes sure git guidelines and good practices are followed
Udit Parihar	Backend Developer	Backend Development
Ayush Kaushik	Frontend Lead	Review frontend Pull Requests
Ridhima Tigadi	Backend Lead	Review backend Pull Requests

9. Checklist

Checklist for team organisation with DONE/ ON TRACK/ ISSUE status.

Task Item	**Status**
Finalised meeting schedule	DONE
GitHub Master chosen	DONE
Agreed on Tech Stack	DONE
Learning approved tech stack	ON TRACK
Team’s understanding of Milestone 1	DONE
GitHub repository organized	DONE